Mask

Your AI doesn’t need everything.

48% of employees using AI at work say they have uploaded company information to public AI tools.

Source: University of Melbourne & KPMG, Trust, attitudes and use of AI: a global study 2025, pp. 75–76. 47 countries. Self-reported.

REAL OUTPUT FROM OUR LIVE SITE

What Claude actually received

PASTED

C-1042,Emma Wilson,emma.wilson@example.com,“I was charged twice for the same order.…”

C-2087,Oliver Davis,oliver.davis@example.com,“…still haven’t received my refund.…”

C-6074,Lucas Harris,lucas.harris@example.com,“Hi, Lucas Harris here. I’ve been billed twice…”

CLAUDE RECEIVED

[ACCOUNT_ID],**[PERSON_1]**,**[EMAIL]**,“I was charged twice for the same order.…”

C-2087,**[PERSON_2]**,**[EMAIL]**,“…still haven’t received my refund.…”

[ACCOUNT_ID],**[PERSON_6]**,**[EMAIL]**,“Hi, **[PERSON_6]** here. I’ve been billed twice…”

Names and emails hidden, complaints untouched. One known gap: C-2087 went through, because our rules read it as a possible business number.

3 of 6 rows shown, shortened. Captured 8 Nov 2026, fictional data. Try it yourself: mask-local-demo.netlify.app

DETECT · SELECT · TRANSFORM

One protection per data type

Data type	Protection	Example
Person names	**Pseudonymised**: same person, same label. The key stays on your device.	Emma Wilson → [PERSON_1]
Emails, IDs, secrets	**Typed placeholder**, checked by validator rules such as the IBAN checksum	emma.wilson@… → [EMAIL]
Passages the question doesn’t need	**Removed**	Next slide
Facts the question needs	**Kept** word for word	“charged twice”

No model rewrites your text: edits happen at exact character positions.

SAME NOTE, TWO REQUESTS

The question decides what goes

EMAIL TO MY MANAGER

Fit for desk duties from 18 November. No lifting over 5 kg.

~~Bipolar II, diagnosed in 2021. Lamotrigine 100 mg.~~

Mental-health history removed

SUMMARY FOR MY NEW GP

Fit for desk duties from 18 November. No lifting over 5 kg.

Bipolar II, diagnosed in 2021. Lamotrigine 100 mg.

Kept: the GP needs it

Shortened from real Mask2 outputs (local Python pipeline, not yet the browser). Synthetic note; the name is hidden in both.

SAFEPROMPT-BENCH · 1,500 DOCUMENTS · ORIGINAL SCORER

Since AI’s own benchmark

Detector	Identifiers F1	Names F1	Clean docs flagged
SafePrompt 0.1.0	98.3	0 (not supported)	1 / 299
Mask1	**98.9**	**100**	**15 / 299**
Mask2 (newest, site default)	98.3	100	57 / 299

The cost: more false alerts. Mask1 flags 15 of 299 clean documents; SafePrompt flags one. Fewer is better.

Synthetic data. Detection scoring only. The benchmark’s names use a few fixed forms, so 100 is not real-world name detection.

HOW MASK MAKES MONEY

Business model

INDIVIDUALS

Free

Browser extension for the chatbots people already use. This is how people find us.

TEAMS

€5 / user / month

The company’s own data rules, an admin view, and a log of what was hidden.

PLATFORMS

Licence

For AI vendors and contact-centre analytics that want to take in customer text safely.

Why the margins work: filtering runs on the user’s own device, so we pay for almost no servers.

Proposed model and starting price, to be tested in a pilot. We have no customers yet.

HONEST STATUS

Built this weekend, and what still fails

Built

Three privacy models, fine-tuned from pretrained encoders on our own synthetic data

Running in the browser, in a public demo anyone can try

Benchmark results and all evidence in our repository

Still fails, on our harder test set

20/58	unneeded clauses removed
8/405	needed facts lost
24/323	identifiers missed

English only. Synthetic data, no human audit yet.

Mask2 on 80 held-out synthetic messages. Pseudonymised text is still personal data: we promise less exposure, not anonymity.

NEXT

01

Support-team pilot

Measure whether answers on protected text stay as good

02

Company rules

Integrations that apply each company’s own data rules

03

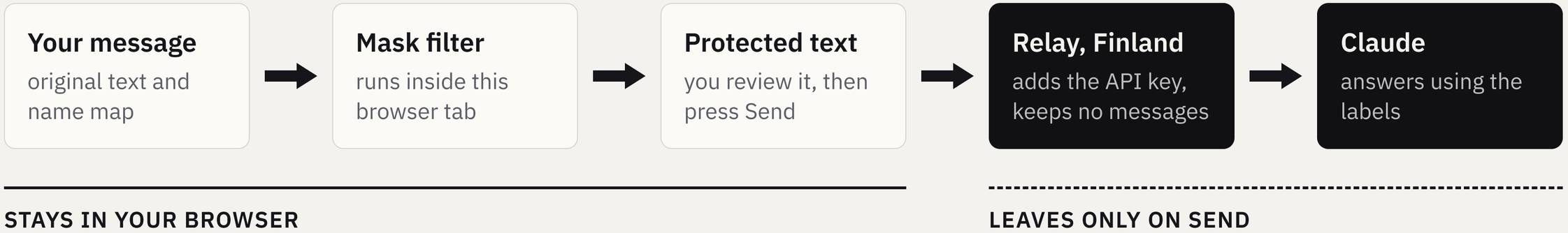
Every AI tool

A browser extension that works inside the chatbot you already use. No separate tab.

Your AI doesn’t need everything.

Mask · mask-local-demo.netlify.app · Next steps are proposals; we have no customers or pilots yet.

How the data flows



The reply comes back with [PERSON_n] labels. Real names are restored in your browser, on request. Emails and IDs stay as placeholders.

Today this is a separate demo page. **The final version** works inside the chatbot you already use, through a browser extension: no separate tab.

ONNX Runtime Web on CPU WebAssembly. The relay checks the site origin, rate-limits and rejects duplicate sends. Source: mask/browser/README.md.

SafePrompt-Bench, all four detectors

Detector	Identifiers F1	Names F1	Secrets F1	Clean docs flagged
SafePrompt 0.1.0	98.3	0.0	100	1 / 299
Mask1 (DeBERTa base)	98.9	100	100	15 / 299
Mask2 (newest base)	98.3	100	100	57 / 299
MaskMini (MiniLM)	98.9	90.3	100	132 / 299

Mask1 and MaskMini score identically on identifiers: that part comes mostly from our shared validator rules. The models add names. Unrounded, Mask2 is 98.28 against SafePrompt’s 98.34.

1,500 synthetic documents, unmodified scorer. Names use fixed forms; secrets follow known formats. Source: evidence/safeprompt-bench-2026-11-08.

Why a model, not just rules

RULES ONLY

8/30

identifiers caught

MASK1: MODEL AND RULES

30/30

identifiers caught, and 182 of 182 needed facts kept

Rules handle anything with a checkable format: IBANs, cards, 25 national ID schemes. Names and relevance have no format, so they need a model.

Original development set, 50 synthetic messages. Source: evidence/privacy-v1/section-training-20261108/COMPARISON.md.